P1 — MASTER provenance: the symbols of PROJECT spacetimes

Which symbol denotes which object in every original file, where the files contradict one another, and the unique symbols that the project uses instead

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0. Purpose, and the result in one page

Hard Rule 5 of the task specification says that the original files of this project often use the *same* symbol for completely *different* objects, that this problem must be handled completely and without ambiguity, and that the proof of the solution is a MASTER provenance: a clear account of the symbol identifications and of the remedies, for each and every relevant file, followed by the consistent use of unique symbols throughout the project. This document is that provenance.

The original files are

- the 14 Mathematica notebooks of the project folder (identifiers N01 to N14; Sec. 5.1),
- nine published papers (D01 to D09; Sec. 5.2), and
- the files of the author's earlier project NCSS, which the task specification asks to be read for context (Sec. 5.3).

The task specification calls the original files "yours and mine": some were written by the author, some were generated for him (the notebook `7+1.nb`, for example, says of itself that a script produced it). This document makes no difference between them.

The problem, measured. A program written for this purpose finds 1921 different symbols that the notebooks define. 350 of them are defined in two or more notebooks, and for 76 of these the defining statements differ from notebook to notebook (Appendix A lists all of them). The papers and the project NCSS add collisions of their own: the index of the 8×8 matrices τ^A runs from 1 to 8 in the papers D07 and D08 and from 0 to 7 in the notebooks; the "strength of the force" D of NCSS is one half of the quantity that carries this letter here.

The remedy (Sec. 2) is one rule with three parts.

1. Every object that the project uses has exactly one symbol. The symbols are listed in a registry, `provenance/symbols.csv` (113 entries; Sec. 3).
2. A symbol of the project cannot be mistaken for a symbol of an original file. Every name that the programs and data files use for a matrix or a field carries the tag **1644** (4+4 spacetime) or **1671** (7+1 spacetime), and every name that belongs to the new scalar field carries the suffix **VS**. No symbol of any notebook contains such a tag or suffix.
3. An object taken from an original file is identified once, here and in the document that introduces it, by *file and original symbol*; after that only the symbol of the project is used.

The proof (Sec. 8) is a set of 9 machine checks, all of which pass. They establish that the registry has no duplicate, that none of its 78 names is a symbol of any notebook, that every tagged name occurring in any data file or document of the project is registered, that the manuscripts contain none of the original names of the objects that were taken over, and that every plain letter of the registry that coincides with a symbol of a notebook is listed with both meanings.

What a reader should take away. If a symbol carries a tag or the suffix VS, look it up in Sec. 3: it has one meaning, in every file of the project. If a formula of an *original* file is to be compared with the project, look up the file in Sec. 5 and the symbol in Sec. 6: the same letter may denote something else there.

0.1 How to reproduce the lists and the checks

From the root of the repository, with the notebooks in that folder (they are not part of the repository; `provenance/inputs_sha256.txt` pins them, and the first script refuses to run if a checksum differs):

```
python3 tools/scan_notebook_symbols.py         # symbols of the notebooks -> provenance/generated/
python3 code/scripts/write_symbol_registry.py   # the registry             ->
↪  provenance/symbols.csv
python3 code/scripts/check_symbols.py          # the checks              -> data/symbol_checks.txt
python3 code/scripts/symbol_concordance.py      # letters in the documents ->
↪  data/symbol_concordance.csv
python3 code/scripts/assemble_docs.py docs_src/P1_master_symbols.src.md
bash code/scripts/build_docs.sh provenance/P1_master_symbols.md
```

The scripts need only Python 3. The first converts each notebook to text with `tools/nb2txt.py` and takes about fifteen seconds; what it counts as "a notebook defines a symbol" is stated in its header and in Appendix A.

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1. The problem

Four examples show the kinds of collision that occur. Each is treated in Secs. 5 and 6.

The same name for different matrices. In `4+4.nb` (N01) the name `γ` is a list of sixteen real 4×4 matrices, a basis of all real 4×4 matrices. In `Dirac_Nash_Identity.nb` (N05) `γ` is a list of four 4×4 matrices. In `Generalized _Kronecker_Delta_4+4.nb` (N09) `γ` is a tensor index. In the paper D06, γ^{AB} are fifteen real 4×4 matrices labelled by a pair of indices with six values. In this project γ denotes the eight real 16×16 Dirac matrices of an eight-dimensional spacetime, and there are two different sets of them.

The same name for different coordinates. Nine notebooks define a list `X` of coordinates. In most of them it is $\{x0, \dots, x7\}$ with $x0$ a hidden space-like coordinate and $x4$ the time. In `7+1.nb` (N02) the same eight names are used, but $x0$ is the time. In `Dirac_Nash_Identity.nb` (N05) `X` has six entries, $x1$ to $x6$. `Einstein-Lovelock-Eternal.nb` (N08) uses the coordinates $x1$ to $x8$, and there $x8$ plays the part that $x0$ plays elsewhere.

Two objects under almost the same name in one file. `7+1.nb` (N02) contains two sets of Dirac matrices for the 7+1 signature, the real set `Γ17` and the complex set `Γ71`; two metrics, `η17` and `η71` $= -$`η17`; and two chirality matrices, `χ17` and `χ71`. A reader who exchanges the two digits obtains a different theory.

The same letter for quantities that differ by a factor. The project NCSS calls $D = \alpha m_\nu / 2 m_\chi$ the strength of the force between two fermions; two fermions bind for $D > 0.83990$. Here the corresponding number is $\alpha_{VS} m / m_s$, without the factor $\frac{1}{2}$, and two fermions bind when it exceeds 1.67981. A reader who carries a value of D from one project to the other is wrong by a factor of two.

The project itself made errors of this kind before the rule of Sec. 2 was enforced by a program. They are recorded in Sec. 7 and in the error log (`provenance/P9_error_log`, entries E7, E35, E41 and E42).

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2. The remedy

Rule 1: one object, one symbol; one symbol, one object. The registry of Sec. 3 is the authority. A new object gets a new entry before it is used in a document.

Rule 2: tags. The tag **1644** marks an object of the 4+4 spacetime, the spacetime of the field dirac1644 (16 components; signature 4+4). The tag **1671** marks an object of the 7+1 spacetime, the spacetime of the field dirac1671. In formulas the tag is a subscript, γ^A_{1644}; in programs and data files it is the end of the name, `gamma1644`. The suffix **VS** marks the quantities that define the Vorton-Scalar: Φ_{VS}, `PhiVS`.

Rule 3: identification by file and original symbol. When an object is taken from an original file, the document that introduces it states the file and the original symbol, for example "γ^A_{1644} is `T16^A[A]` of `4+4.nb`", and a machine check compares the two entry by entry where that is possible (checks `44-N1` to `44-N3`, `44-P3`, `71-N1` to `71-N3` of the program `spacetimes matrices`, which read the matrices that Mathematica printed in the notebooks). The original symbol is not used again.

Rule 4: the scope of a dropped tag. Inside a section that deals with one spacetime only, the tag may be dropped in derivations; the equations that define an object or state a result carry it. In the sections that the two manuscripts share, β, α^j, χ, `h` and m denote the matrices and the mass of the spacetime at hand (Sec. 3.4). Two symbols of this kind deserve a warning, because they are the same letter for corresponding but different matrices: in the 7+1 spacetime $\chi \equiv \chi_{1671} = i\,\omega_{1671}$ is complex, in the 4+4 spacetime $\chi \equiv \omega_{1644}$ is real; and the adjoint $\bar\psi \equiv \psi^\dagger\beta$ of the quantized theories differs, in 4+4, from the covariant adjoint $\tilde\psi \equiv \psi^\dagger C^-_{1644}$.

Rule 5: letters without a tag. A document of physics cannot give every quantity a letter of its own; the alphabet is too small. Letters that the project uses for more than one quantity are listed in Sec. 4, with the subscript, argument or section that tells the meanings apart. Plain letters of the project that coincide with a symbol of a notebook are listed, with both meanings, in Sec. 6.3; a machine check (`S9`) guarantees that this list is complete for the symbols of the registry.

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3. The registry: the symbols of the project

The tables of this section are generated from `provenance/symbols.csv`. The column "Name in files" gives the names used in the output of the programs and in the data files; "Defined in" uses the abbreviations DM for the provenance "dirac matrices", S1 to S4 for the supplements and M for the two manuscripts (the section numbers of the manuscripts are the same in both). "Origin" names the notebook (N01 to N14) or the paper (D01 to D09) and the original symbol; "new" means that the project introduced the object. The word NOT marks an original symbol that is easily confused with the right one.

3.1 Coordinates and indices

Symbol	Name in files	Meaning	Defined in	Origin
$X = \{x0, \dots, x7\}$	`x0` `x1` `x2` `x3` `x4` `x5` `x6` `x7`	the eight coordinates of both spacetimes, never renamed: $x0$ hidden space-like; $x1, x2, x3$ ordinary space; $x4$ ordinary time; $x5, x6, x7$ additional coordinates (time-like in 4+4, space-like in 7+1)	DM 0.1; M I	N01 (text at its head) and N04 `X` (section DEFCHART): the same eight names with the same meaning. N02 uses the same eight names with `x0` as the time: related by swap04

Symbol	Name in files	Meaning	Defined in	Origin
swap_{04}	`swap04`	the exchange of the index values 0 and 4; it carries the labels of `7+1.nb` (time at index 0) into those of the project (time at index 4)	DM 0.1, 2.1	new
A, B, C, D	—	index with the values $0, \ldots, 7$; always labels the coordinate xA	DM 0.1; M I	new
a, b	—	in S1 to S4 and in the manuscripts: index of the four extra axes, values 0, 5, 6, 7. In DM Sec. 1.8.4 and in the texts of the checks `44-D6`, `44-D7`: the four space-like axes of 4+4, values 0, 1, 2, 3 (stated there)	M I; S1 0.1; DM 1.8.4	new
t (index)	—	index of the three additional coordinates, values 5, 6, 7	DM 1.8.4; M I	new
i, j	—	index of space-like coordinates; values 1, 2, 3 (ordinary space) unless a sum is said to run over all space-like coordinates of the spacetime	M I	new
h (index)	—	index with the values 1, 2, 3 that labels the 4×4 matrices s_h, t_h	DM 1.1; M II	N01, N02, N05 `h` in `s4by4[h]`, `t4by4[h]`

3.2 Matrices of the 4+4 spacetime

Symbol	Name in files	Meaning	Defined in	Origin
η_{1644}	`eta1644`	metric of the 4+4 spacetime, $\mathrm{diag}(+1,+1,+1,+1,-1,-1,-1,-1)$: space-like $= +1$	DM 1.4, Eq. (1.3)	N01, N02, N05, N10–N12 `η4488`; in N01 also `η8` and `g_AB`; D08 G_{AB} with the index value 8 in the place of the value 0
s_h, t_h	—	the three self-dual and the three anti-self-dual real antisymmetric 4×4 matrices	DM 1.1; M II (4+4), App. A (7+1)	N01, N02, N05, N10–N12 `s4by4[h]`, `t4by4[h]`; the letters are those of D06 (s^h self-dual, t^h anti-self-dual)
τ^A_{1644}, $\bar{\tau}^A_{1644}$	`tau1644` `taubar1644`	the real 8×8 blocks of the 4+4 Dirac matrices (reduced generators of the split octonion algebra)	DM 1.1, App. B; M II	N01, N10–N12 `τ[A]` and `τ` with an overbar; N02 `τ[A]`, `τbar[A]`; matrices of this kind are the τ^A, $\bar{\tau}^A$ of D07 and D08, whose index runs from 1 to 8 (the value 8 there is the value 0 here)
σ_{88}	`sigma88`	the 8×8 matrix with the 4×4 blocks $(0, 1_4; 1_4, 0)$	DM 1.1	N01 `σ`, `σ88`; N02 `σ`; N05 `σ88`
γ^A_{1644}	`gamma1644`	the eight real 16×16 Dirac matrices of the 4+4 spacetime	DM 1.1, 1.2; M II	N01, N10–N12 `T16^A[A]`; N02 `T16[A]`; the t^A of D08 are built in the same way
Σ^{AB}_{1644}	`Sigma1644`	$\frac{1}{4}[\gamma^A_{1644}, \gamma^B_{1644}]$, the scaled commutators; they span $\mathfrak{so}(4,4)$	DM 1.5; M II	new name; the same matrices are `SAB` and `sAB[A][B]` of N10–N12; NOT `sAB` of N02, which is a sign
ω_{1644}	`omega1644`	the product $\gamma^0_{1644} \cdots \gamma^7_{1644}$; real chirality matrix of 4+4, $\mathrm{diag}(-1_8, +1_8)$	DM 1.6; M II	N01 `T16^A[8]`; N02 `T16[8]`
P^L_{1644}, P^R_{1644}	`PL1644` `PR1644`	the real chiral projectors $\frac{1}{2}(1 \mp \omega_{1644})$	DM 1.6; M II	N01 `P_L`, `P_R`; N02 `PL44`, `PR44`

Symbol	Name in files	Meaning	Defined in	Origin
C^{-}_{1644}, C^{+}_{1644}	`Cm1644` `Cp1644`	the products $\gamma^0\gamma^1\gamma^2\gamma^3$ and $\gamma^4\gamma^5\gamma^6\gamma^7$ of the 4+4 matrices; $C^{\mp}\gamma^A(C^{\mp})^{-1} = \mp(\gamma^A)^T$	DM 1.7; M II	N01, N02, N10–N12 `σ16` (C^-); N02 `S44` (C^+)
W_{1644}	`W1644`	$i\,C^{-}_{1644}\gamma^4_{1644}$, the grading: Hermitian, $W^2 = 1$; density of the covariant current	DM 1.8.4; M III	new
K_{1644}	`K1644`	generator of the evolution of dirac1644 in $x4$: $\partial_4\psi = K\psi$	DM 1.8.4, Eq. (1.15); M III	new
β_{1644}, α^j_{1644}	`beta1644` `alpha1644`	the Hermitian matrices $-i\gamma^4_{1644}$ and $-\gamma^4_{1644}\gamma^j_{1644}$ ($j = 0, 1, 2, 3$) of the quantized 4+4 theory	S1 2; M III	new
h_{1644}	`h1644`	Hamiltonian matrix of the quantized 4+4 theory, iK_{1644} for fields independent of $x5, x6, x7$	DM 1.8.4; M III	new
Π_{1644}	`Pi1644`	the projector $(i\gamma^A_{1644}k_A + m_{1644})/2m_{1644}$ on the plane-wave solutions of the worked example	DM 1.8.1; M III	new

3.3 Matrices of the 7+1 spacetime

Symbol	Name in files	Meaning	Defined in	Origin
η_{1671}	`eta1671`	metric of the 7+1 spacetime, $\mathrm{diag}(-1,-1,-1,-1,+1,-1,-1,-1)$: time-like $= +1$, time at index 4	DM 2.4, Eq. (2.3); M II	N02 `η17` with the index values 0 and 4 exchanged; NOT `η71`
τ^A_{1671}	`tau1671`	the real 8×8 matrices of the 7+1 construction: $\tau^4_{1671} = 1_8$, the other seven represent the imaginary octonion units	DM 2.1, App. B; M II	N02 `τO[swap04(A)]`
γ^A_{1671}	`gamma1671`	the eight real 16×16 Dirac matrices of the 7+1 spacetime (contravariant index)	DM 2.1, 2.2; M II	N02 `Γ17up[swap04(A)]`; NOT `Γ71[A]`
Σ^{AB}_{1671}	`Sigma1671`	$\frac{1}{4}[\gamma^A_{1671}, \gamma^B_{1671}]$, the scaled commutators; they span $\mathfrak{so}(7,1)$	DM 2.5; M II	N02 `Σ17[A,B]`, built from the `Γ17` with lower index: $\Sigma^{AB}_{1671} = \eta^{AA}_{1671}\eta^{BB}_{1671}$ `Σ17[A',B']` with $A' = \mathrm{swap}_{04}(A)$, $B' = \mathrm{swap}_{04}(B)$
ω_{1671}	`omega1671`	the product $\gamma^0_{1671}\cdots\gamma^7_{1671}$; real, $\omega^2_{1671} = -1$	DM 2.6; M II	N02 `Γ17[8]`, `ω17`
χ_{1671}	`chi1671`	$i\,\omega_{1671}$, the (complex) chirality matrix of 7+1	DM 2.6; M II	N02 `χ17`; NOT `χ71`
P^L_{1671}, P^R_{1671}	`PL1671` `PR1671`	the complex chiral projectors $\frac{1}{2}(1 \mp \chi_{1671})$	DM 2.6; M II	N02 `PL17`, `PR17`; NOT `PL71`, `PR71`
U^c_{1671}	`Uc1671`	the unitary matrix $(1 - i\gamma^4_{1671})/\sqrt{2}$ that carries the complex projectors of 7+1 into the real projectors of 4+4	DM 2.6, Eq. (2.5); M II	new
C^{+}_{1671}, C^{-}_{1671}	`Cp1671` `Cm1671`	γ^4_{1671} and $\gamma^4_{1671}\omega_{1671}$; $C^{\pm}\gamma^A(C^{\pm})^{-1} = \pm(\gamma^A)^T$	DM 2.7; M II	new
β_{1671}, α^j_{1671}	`beta1671` `alpha1671`	the real symmetric matrices γ^4_{1671} and $\gamma^4_{1671}\gamma^j_{1671}$ (seven space indices j) of the Hamiltonian	DM 2.8.2; M II	N02 `βD`, `αD[swap04(j)]`
h_{1671}	`h1671`	Hamiltonian matrix of the 7+1 theory, $\alpha^j_{1671}p^j + \beta_{1671}m_{1671}$	DM 2.8.2; M III	N02 `h`
Π_{1671}	`Pi1671`	the projector $(\gamma^A_{1671}k_A + m_{1671})/2m_{1671}$ of the worked example	DM 2.8.1	new

Symbol	Name in files	Meaning	Defined in	Origin
E^L_{1671}, E^R_{1671}	`EL1671` `ER1671`	the chiral frames: 16×8 matrices whose columns span the two eigenspaces of χ_{1671}	DM 2.8.3; M III	N02 `EL`, `ER`
M_{1671}, U^+_{1671}, V^-_{1671}	`M1671` `Upos1671` `Vneg1671`	the 8×8 matrix $m_{1671} + p^j \tau^j_{1671}$ and the matrices of positive- and negative-energy spinors of the worked example (check `71-D8`)	DM 2.8.2	N02 `M` of its Sec. 7.4 (in N02 the letter `M` also denotes other matrices and a mass); the two spinor matrices are new names

The two sets of Dirac matrices are related exactly (check `71-B4`): $\gamma^4_{1671} = \gamma^0_{1644}$; $\gamma^j_{1671} = -\gamma^0_{1644}\gamma^j_{1644}$ for $j = 1,2,3$; $\gamma^0_{1671} = +\gamma^0_{1644}\gamma^5_{1644}\gamma^6_{1644}\gamma^7_{1644}$, $\gamma^5_{1671} = -\gamma^0_{1644}\gamma^4_{1644}\gamma^6_{1644}\gamma^7_{1644}$, $\gamma^6_{1671} = +\gamma^0_{1644}\gamma^4_{1644}\gamma^5_{1644}\gamma^7_{1644}$, $\gamma^7_{1671} = -\gamma^0_{1644}\gamma^4_{1644}\gamma^5_{1644}\gamma^6_{1644}$. A formula that is true for one set is therefore in general false for the other, and the tag is the only thing that tells them apart.

3.4 Fields and their equations

Symbol	Name in files	Meaning	Defined in	Origin
ψ_{1644}	`dirac1644`	the field dirac1644: 16 complex Grassmann components on the 4+4 spacetime; $(\gamma^A_{1644}\partial_A - m_{1644})\psi_{1644} = 0$	DM 1.8; M III	new; `Ψ16` of N01 and N10–N12 is a column of 16 real quantities (in N10–N12 functions of `x0`, `x4` called the wave function of the universe), not this field
ψ^L_{1644}, ψ^R_{1644}	`psiL1644` `psiR1644`	the two 8-component chiral parts of dirac1644 (upper and lower eight components)	DM 1.8.5; M III	new
m_{1644}	`m1644`	mass of the field dirac1644	DM 1.8.1	new
$\tilde{\psi}$	—	the covariant adjoint $\psi^\dagger C^-_{1644}$ of the 4+4 theory (invariant under $\mathrm{Spin}_0(4,4)$)	DM 1.8.2; M III (4+4)	new
j^A_{1644}	`j1644`	the conserved current $i\,\tilde{\psi}\gamma^A_{1644}\psi$ of the covariant 4+4 theory (fermion number)	DM 1.8.2, Eq. (1.13); M III	new
ψ_{1671}	`dirac1671`	the field dirac1671: 16 complex Grassmann components on the 7+1 spacetime; $(i\gamma^A_{1671}\partial_A - m_{1671})\psi_{1671} = 0$	DM 2.8; M III	N02 `dirac16complex`
ϕ^L_{1671}, ϕ^R_{1671}	`phiL1671` `phiR1671`	the two 8-component chiral parts of dirac1671	DM 2.8.3; M III	N02 `φL`, `φR`
m_{1671}	`m1671`	mass of the field dirac1671	DM 2.8.1	N02 `m`, `mD`
$\bar{\psi}$	—	the adjoint $\psi^\dagger\beta$ of the quantized theories ($\beta = \beta_{1671}$ or β_{1644}); in 4+4 it differs from $\tilde{\psi}$	S1 1.2, 3.2; M V	new
β, α^j, χ, `h`, m	—	in the sections common to both spacetimes: the matrices and the mass of the spacetime at hand, i.e. the tagged symbols with the tag dropped; $\chi \equiv \chi_{1671}$ in 7+1 and $\chi \equiv \omega_{1644}$ in 4+4	S1 2, Eq. (2.1); M V (4+4), Eq. for χ	new
$U(\vec{p})$	—	the 16×8 matrix of orthonormal positive-energy spinors at momentum $\vec{p}$	S1 4.1; M VI	new

3.5 Symmetries, pairs and the order parameter

Symbol	Name in files	Meaning	Defined in	Origin
$U(1)_F$	—	the phase symmetry $\psi \to e^{i\epsilon}\psi$: fermion number	S1 1.2; M V	new
$U(1)_{VS}$	`UVS`	the chiral symmetry $\psi \to e^{i\epsilon\chi}\psi$, $\Phi_{VS} \to e^{-2i\epsilon}\Phi_{VS}$, broken by the vacuum value of the Vorton-Scalar	S1 1.2; M V	new
$SU(2)_A$, $SU(2)_B$	—	the two factors of the internal symmetry of the fields that do not depend on the four extra coordinates (zero-mode sector), in both spacetimes; $SO(4) = [SU(2)_A \times SU(2)_B]/Z_2$	S1 2.1; S4 2.4; M V	new
$SU(2)_X$, $SU(2)_Y$	—	the two factors of the rotation group of the four space coordinates $x0, \dots, x3$ of the 4+4 spacetime (NOT the internal $SU(2)_A \times SU(2)_B$)	DM 3.2, App. A	new
$SO(3)_T$, $SU(2)_T$	—	the rotations of the three additional times $x5, x6, x7$ of 4+4, and its covering group; the diagonal subgroup of $SU(2)_A \times SU(2)_B$	S1 6.3; M V, X	new
$U(1)_E$, P (generator)	—	the second central phase symmetry of the zero-mode sector and its traceless generator, $P^2 = 1$: $P = +1$ on the doublets of $SU(2)_A$, -1 on those of $SU(2)_B$	S4 2.4; M V	new
Δ^A_{1644}, Δ^A_{1671}	`DeltaV1644` `DeltaV1671`	the vector pair operators $\psi^T C^- \gamma^A \psi$ of each spacetime	DM 3; M IV	new
Δ^{AB}_{1671}	`DeltaT1671`	the two-form pair operators $\psi^T C^+ \gamma^A \gamma^B \psi$ of the 7+1 spacetime	DM 3.2	new
Δ^a	—	order parameter: expectation value of the vector pair operator along the extra axis $a \in \{0, 5, 6, 7\}$; $\lvert\Delta\rvert^2 \equiv \sum_a \lvert\Delta^a\rvert^2$ (called S in S1 and S4)	S1 6.1; M VIII	new
$\Phi_X(\vec{p})$	—	two-particle content $U(\vec{p})^T X U(-\vec{p})$ of the pair operator $\psi^T X \psi$	S1 4.1; M VI	new
`M`	—	the 8×8 gap matrix on the internal states of a pair	S1 6.1; M VIII	new
$d_\pm$	—	the two fourfold gaps of a general state, $d_\pm^2 = \lvert\Delta\rvert^2 \pm \sqrt{\lvert\Delta\rvert^4 - \lvert\eta_{ab}\Delta^a\Delta^b\rvert^2}$	S4 4.2; M VIII	new
n^a	—	real unit vector: direction of the order parameter in the space of the four extra axes	S1 6.3; M VIII	new

3.6 The Vorton-Scalar

Symbol	Name in files	Meaning	Defined in	Origin
Φ_{VS}	`PhiVS`	the Vorton-Scalar: a complex scalar field	S1 1.2; M V	new (the name is that of the author's project NCSS, where the scalar is real)
v_{VS}	`vVS`	vacuum value: $\langle\Phi_{VS}\rangle = v_{VS}/\sqrt{2}$	S1 1.2; M V	new
λ_{VS}	`lambdaVS` `lamVS`	self-coupling of the Vorton-Scalar	S1 1.2; M V	new
y_{VS}	`yVS`	Yukawa coupling of the Vorton-Scalar to the fermion field	S1 1.2; M V	new
s_{VS}	`sVS`	the radial mode of the Vorton-Scalar (real field)	S1 1.4; M V	new
m_s	`msVS`	mass of the radial mode, $\sqrt{2\lambda_{VS}}\, v_{VS}$	S1 1.4; M V	new

Symbol	Name in files	Meaning	Defined in	Origin
ϑ_{VS}	`thetaVS` `thVS`	the phase mode of the Vorton-Scalar (real field, massless)	S1 1.4; M V	new
g_{VS}	`gVS`	coupling of both modes to the fermions, $m/v_{VS} = y_{VS}/\sqrt{2}$	S1 1.4; M V	new
α_{VS}	`alphaVS` `aVS`	$g_{VS}^2/4\pi$	S1 1.4; M V	new
D_{VS}	`DVS`	$\alpha_{VS}\, m/m_s$: two fermions bind if and only if $D_{VS} > D_c$	S1 5.3; M VII	new
J^A_{VS}	`JVS`	the neutral current of the Vorton-Scalar (bilinear in Φ_{VS})	S1 1.5; M V	new
J^A_5	`J5`	the axial current $\bar\psi\gamma^A\chi\psi$ of the fermion field	S1 1.5; M V	new
G_{VS}	`GVS`	constant of the Fermi type of the neutral-current interaction, $G_{VS}/\sqrt{2} = 1/(2v_{VS}^2)$	S1 1.5; M V	new
$\mathcal{L}^{VS}_{1644}$, $\mathcal{L}^{VS}_{1671}$	`LVS`	the Lagrangians that define the Vorton-Scalar in the two spacetimes	S1 1.2, 3.2; M V	new
C (coupling)	—	$g_{VS}^2 m^2/m_s^2 = g_{VS}^4/2\lambda_{VS}$, the one number on which the fermion mass in matter depends	S4 5.1; M IX	new

3.7 Pairing, the gap equation and the effective action of Cornwall, Jackiw and Tomboulis

Symbol	Name in files	Meaning	Defined in	Origin
ν	—	the number of internal states of a particle of given momentum, $\nu = 8$	S1 5; M VII	new
F_S, F_P	`F_scalar` `F_pseudoscalar`	exchange amplitudes of the radial and of the phase mode in a pair channel	S1 4.1; M VI	new
V_{NC}, $V_s(r)$	—	contact interaction of the neutral current in the hidden-axis channel; Yukawa potential of the radial mode	S1 4.2; M VI	new
σ_X	—	the sign $+1$ (vector pairs along the extra axes) or -1 (their two-form pairs) of the phase-mode term of the kernel	M VII	new
$W(k,k')$	—	s-wave average of the exchange attraction: kernel of the gap equation	S1 5.1; M VII	new
$\Delta(k)$, Δ_F	—	gap function; its value at the Fermi momentum	S1 5.1; M VII	new
ξ_k, E_k, $\kappa(k)$	—	$\sqrt{m^2+k^2}-\mu$; quasi-particle energy $\sqrt{\xi_k^2+\Delta(k)^2}$; pair correlation $\langle b(-\vec{k})b(\vec{k})\rangle$	S4 3.1; M VII	new
k_F, E_F, μ	—	Fermi momentum, $n = \nu k_F^3/6\pi^2$; Fermi kinetic energy $\sqrt{m^2+k_F^2}-m$; chemical potential	S1 5; M VII	new
$N(0)$	—	density of states at the Fermi surface per internal state, $k_F\sqrt{m^2+k_F^2}/2\pi^2$	S1 5.2; M VII	new
λ, λ_S, λ_P	—	dimensionless pairing coupling $N(0)W(k_F,k_F)$ and its parts from the radial and the phase mode	S1 5.2; M VII	new
T_c	—	critical temperature of the paired state	S1 5.2; M VII	new
D_c	—	critical value of D_{VS} for a bound pair in the Yukawa potential	S1 5.3; M VII	new

Symbol	Name in files	Meaning	Defined in	Origin
a_{sc}, a_{BB}	—	scattering length of two fermions (written a in S1, S3, S4); scattering length of two bound pairs	S1 5.3; M VII, VIII	new
$\Gamma[\phi, G]$, Γ_2	—	the effective action for composite operators of Cornwall, Jackiw and Tomboulis; its two-particle-irreducible part	S4 1; M VII	new (method of the cited paper)
$\mathcal{S}$, D_s, D_ϑ	—	propagator of the doubled fermion field; propagators of the radial and of the phase mode	S4 2.1; M VII	new
$V[\Delta]$	—	the CJT effective potential of the paired state, measured from the unpaired state	S4 3.1; M VII	new
$\eta(T)$	—	largest eigenvalue of the linearized gap equation, $\eta(T_c) = 1$	S4 3.4; M VII	new
$A_2(T)$, A_4	—	coefficients of the expansion of V along a gap function of fixed shape (written α, β in S4)	S4 3.4; M VII	new
$a(T)$, b, K_{GL}	—	Ginzburg-Landau coefficients of the weak-coupling free energy (K_{GL} is written K in S1 and S4)	S1 6.2; M VIII	new
c_s	—	speed of first sound of the superfluid, $c_s^2 = dP/d\rho$	S1 6.4; M VIII	new

3.8 Matter, temperature and vacuum

Symbol	Name in files	Meaning	Defined in	Origin
ϕ	—	expectation value of the Vorton-Scalar, $\langle \Phi_{VS} \rangle = \phi/\sqrt{2}$; the fermion mass is $g_{VS}\phi$	S4 2.1; M VII	new
m^*, x	—	fermion mass in the medium, and $x = m^*/m$ (in the manuscripts Sec. XII A, x denotes k_F/m, stated there)	S4 5.1; M IX	new
$\varepsilon(k_F, x)$, $V_{\mathrm{sea}}(x)$	—	energy density of the unpaired state in units of m^4; renormalized energy of the filled negative-energy states	S4 5.1; M IX	new
B	—	energy density of the state without expectation value of the Vorton-Scalar (bag constant)	S4 5.2; M IX	new
E_x	—	exchange (Fock) energy of the two modes	S4 5.4; M IX	new
M_s, M_ϑ, a_s, a_ϑ	—	thermal masses of the two boson modes in the Hartree approximation, and the curvatures U''_{eff}, U'_{eff}/ϕ	S4 6; M IX	new
$P(M)$, $Q(M)$	—	thermal part of the boson loop and of $\frac{1}{2}\mathrm{Tr}\ln$	S4 6; M IX	new
T_{high}, T_N, $T_{1/2}$, T_0	—	two closed estimates of the temperature at which the fermion mass vanishes; temperatures at which the expectation value has fallen to half and to zero	S4 6; M IX	new
$\sigma(p)$	—	self-energy that a pair condensate in the vacuum gives to the fermions: mass splitting $m \pm \sigma$	S4 7.2; M IX	new
α_c, Λ (cutoff)	—	critical value of α_{VS} for a pair condensate in the vacuum; momentum cutoff of the ladder equation	S4 7.2; M IX	new

3.9 Cosmology

Symbol	Name in files	Meaning	Defined in	Origin
N	—	density of the dark radiation today in units of the density of one massless neutrino species	S3 3.2; M XI	new
z_t, T_d	—	transition redshift of the stepped dark fluid, $m = T_{d0}(1+z_t)$; temperature of the dark sector	S3 3.2; M XI, D	new
r_{step}	—	factor by which the density of the dark radiation rises when the fermions annihilate, $15^{1/3}$ for this model	S3 1; M XI	new
ω_b, ω_c, h, H_0, A_s, n_s, τ	—	the six parameters of the standard cosmological model: densities of baryons and of cold dark matter, Hubble constant $H_0 = 100\,h$ km s^{-1} Mpc^{-1}, amplitude and tilt of the primordial spectrum, optical depth	S3 2.1; M XI	standard notation
$\Delta\chi^2$	—	worsening of the best fit to the mock data relative to the standard model	S3 3.3; M XI	new
θ_s, r_s, D_A	—	angular size of the sound horizon; sound horizon; distance to last scattering	S3 3.1; M XI	standard notation
c_{s0}	—	sound speed of superfluid dark matter today at the mean cosmic density	S3 4.1; M XII	new
ρ_0, R_{TF}	—	constant of the equation of state of a condensate of bound pairs; Thomas-Fermi radius of a core supported by its pressure	S3 4.1; M XII	new
m_{deg}, $m_{B,\mathrm{max}}$	—	largest fermion mass for degeneracy, and largest boson mass for condensation, at a given density and velocity dispersion	S3 4.3; M XII	new

4. Letters that have more than one meaning in the documents of the project

The tagged symbols of Sec. 3 have one meaning each. The plain letters below are used for more than one quantity. In every case the meanings are told apart by a subscript, a superscript, an argument, a different typeface, or by the section in which the letter occurs; the last column says which. The table covers the two manuscripts (M, with the Roman number of the section), the four supplements and “dirac matrices” (DM).

The table was written by hand and then checked against a list made by a program: `data/symbol_concordance.csv` (script `code/scripts/symbol_concordance.py`) gives, for every Latin and Greek letter and every document, each form in which the letter occurs in a formula, with the number of occurrences by section. A reader who wants to see every use of a letter finds it there.

Letter	Meanings	How they are told apart
a, b	index of the four extra axes, values 0, 5, 6, 7 (S1 to S4; M IV to XIII); index of the sixteen components of the field, ψ_a (M III, IV); in DM Sec. 1.8.4, index of the four space-like axes of 4+4; scale factor a of the universe (M XII); Ginzburg-Landau coefficients $a(T)$, b; curvatures a_s, a_ϑ of the potential; scattering length (S1, S3, S4); annihilation operators $b(\vec{p})$	an index is a subscript or superscript, and its range is stated where it is used; $a(T)$ carries its argument, a_s, a_ϑ their subscript; the manuscripts write the scattering length a_{sc}; an operator carries its argument
A, B	indices with the values 0 to 7; the coefficients $A_2(T)$, A_4; the amplitude A_s of the primordial spectrum; the bag constant B (S4; M IX); the Killing form $B(X,Y)$ (DM; M II); subscripts of the groups $SU(2)_A$, $SU(2)_B$ and of quantities of a bound pair (m_B, n_B, a_{BB})	subscript; argument; position

Letter	Meanings	How they are told apart
C	index; the matrices C^-, C^+; the coupling $C = g_{VS}^2 m^2/m_s^2$; heat capacities C_s, C_n; angular power spectra C_ℓ; the labels C1 to C7 of the conditions on a Higgs field	superscript $\mp$ and tag; subscript; a label is followed by a digit
D	index; D_{VS} and its critical value D_c; inverse propagator $D^{-1}(\phi)$ and the boson propagators D_s, D_ϑ; $D_\ell \equiv \ell(\ell+1)C_\ell/2\pi$; distance D_A	subscript
d	the two gaps $d_\pm$ of a general state; annihilation operators $d_s(\vec{p})$ of antiparticles (M III); differential	subscript and argument
E	energies E, E_k, E_F, E_b, E_x; the chiral frames $E_{1671}^{L,R}$; subscript of the group $U(1)_E$	tag and superscript; position
G	full propagator G of the effective action (M VII, Appendix B; S4 Sec. 1); the constant of gravitation (M XII; S3 Sec. 4.1); G_{VS}; Fermi constant G_F	subscript; section
Γ	effective action $\Gamma[\phi, G]$ and its part Γ_2 (M VII; S4); a vertex matrix Γ in the exchange amplitude (M VI; S1 Sec. 4)	argument or subscript; section
h	index of the matrices s_h, t_h; Hubble constant in units of 100 km s^{-1} Mpc^{-1} (M XI, XII; S3); Hamiltonian matrix h	index position; the matrix is set in sans-serif type and carries a tag where it is defined
K	K_{1644}, generator of the evolution in $x4$ (DM; M III of 4+4); the source $K(x, y)$ of the effective action; gradient coefficient of the Ginzburg-Landau energy (S1, S4)	tag; argument; the manuscripts write the gradient coefficient K_{GL}
M	mass of an exchanged boson (M VI, IX; S4); thermal masses M_s, M_ϑ; gap matrix M; the matrix M_{1671} of a worked example (DM; M III of 7+1); M_W, M_Z; solar mass $M_\odot$	subscript; typeface; tag
N	density of dark radiation (M XI; S3); density of states $N(0)$; norms N_X, N_a of pair amplitudes; N_{eff}; subscript of the temperature T_N	argument; subscript
n	number density; direction n^a of the order parameter; winding number of a vortex (M V; S1); tilt n_s; size of the matrices 1_n and of the spectra in the mock likelihood	index; subscript; section
P	chiral projectors P^L, P^R; the central generator P of $U(1)_E$ (M V; S4); pressure P, $P(\rho)$; thermal boson loops $P(M)$, P_s, P_ϑ; subscript of the amplitude F_P and of the coupling λ_P of the phase mode	superscript and tag; argument; subscript; section
Q	thermal part $Q(M)$ of $\frac{1}{2}\mathrm{Tr}\ln$ (M IX; S4); electric charge (M X; S2)	argument; section
r, s	labels of the eight internal states of a particle, u_s, b_r (M III, VI); rank r of a product of Dirac matrices (DM; M II, IV); radius r; the matrices s_h; the radial mode s_{VS} and the subscript s of its quantities (m_s, D_s, M_s, a_s, P_s); scaling factor s of a trial gap function (M VII; S4); subscript of the superfluid state (S_s, C_s); standard cosmological subscripts (c_s, r_s, θ_s, n_s, A_s); r_{step}, r_{drag}	index position; suffix VS; the quantity that carries the subscript
S	element S of the spin group (DM; M II to IV); classical action $S[\Phi]$; entropy S, S_s, S_n; propagator $\mathcal{S}$ of the doubled fermion field; subscript of F_S, λ_S (radial mode); in S1 and S4, $S = \sum_a \lvert\Delta^a\rvert^2$	argument; subscript; typeface; the manuscripts write $\lvert\Delta\rvert^2$ for the last
Σ	the scaled commutators Σ^{AB}; self-energy Σ, $\Sigma[\phi, G]$ of the effective action (M VII, Appendix B; S4)	indices
T	temperature, with T_c, $T_{1/2}$, T_0, T_{high}, T_N, T_d; transposition (superscript); generators T_A^3, T_B^3 of $SU(2)_A$, $SU(2)_B$ (M X; S2); subscript of the group $SO(3)_T$ of the additional times	position; indices
t	index of the three additional coordinates, values 5, 6, 7; the matrices t_h; subscript of the transition redshift z_t	index position
U	matrix $U(\vec{p})$ of positive-energy spinors; U_{1671}^c; the effective potential U_{eff}; the groups $U(1)$	argument; tag; subscript
u, v	the real spinors u_s, v_s of the plane waves (M III); reduced radial wave function u of the bound pair (M VII; S1); the vacuum value v_{VS} and the electroweak value v_{EW}; Fermi velocity v_F; a vector v in the definition of the Clifford algebra (M II)	subscript; section
V	effective potential V, $V[\Delta]$, $V(\phi, G)$; the interactions V_{NC}, $V_s(r)$, V_{BCS}; V_{sea}; the matrices V_1, V_2 of the exchange amplitude (M VI); volume V of the quantization box (M III) and V_4 of the extra dimensions	subscript; argument; section

Letter	Meanings	How they are told apart
W	W_{1644}, the grading; kernel $W(k,k')$ of the gap equation; generating functional $W[J,K]$; subscript of the weak mixing angle θ_W and of M_W	tag; argument; position
x, y	the coordinates $x0,\dots,x7$ and points x, y of spacetime; the ratio $x = m^*/m$ (M IX; S4); in M XII A, $x = k_F/m$; the Yukawa coupling y_{VS}	the coordinates carry a digit; the two ratios are defined where they are used; suffix VS
α	the matrices α^j; α_{VS} and its critical value α_c; in S4, a coefficient $\alpha(T)$; in the name "Lyman-α"	index; subscript; the manuscripts write $A_2(T)$ for the coefficient
β	the matrix β; in S4, a coefficient	the manuscripts write A_4 for the coefficient
Δ	order parameter Δ^a and gap $\Delta(k)$, Δ_F; the pair operators Δ^A_{1644}, Δ^A_{1671}; differences $\Delta\chi^2$, ΔN_{eff}	a difference is followed by the quantity that differs
ϵ, ε	parameter of a symmetry transformation; permutation symbol ϵ_{hpq4}; energy density $\varepsilon(k_F,x)$; dimensionless binding energy of the pair (M VII E; S1)	indices; argument; section
η	the metrics η_{1644}, η_{1671}, η^{AB}, η_{ab}; eigenvalue $\eta(T)$ of the linearized gap equation	indices or tag; argument
θ, ϑ	rotation parameters θ_{AB}, θ; angle between two momenta (M VII) and angle of a direction of the order parameter (M VIII); angular size θ_s of the sound horizon; weak mixing angle θ_W; the phase mode ϑ_{VS} and the subscript ϑ of its quantities	subscript; suffix VS; section
κ	pair correlation $\kappa(k)$; frequencies κ_t along the additional times of 4+4 (M III of 4+4; DM); the bound $\kappa = 1$ cm^2/g on the self-interaction (M XII; S3)	argument; index; unit
Λ	Lorentz matrix $\Lambda^A{}_B$; momentum cutoff; the cosmological constant in the name ΛCDM	indices; context
λ	λ_{VS}; pairing couplings λ, λ_S, λ_P; the coupling of the zero-dimensional test model (M Appendix B; S4 Sec. 1.7); thermal wavelength (M XII)	suffix VS; subscript; section
μ, ν	chemical potential μ; the number $\nu = 8$ of internal states; in the unit μK	context
ρ	mass density, with ρ_0, ρ_{deg}; the variable $m_s r$ of the bound-pair equation (M VII E; S1 Sec. 5.3); the ratio $M_W^2/M_Z^2\cos^2\theta_W$ (M X; S2); an exponent in the solution of the ladder equation (M IX)	defined where used; each of the last three occurs in one subsection
σ	the matrix σ_{88}; the sign σ_X; self-energy σ, $\sigma(p)$ (M IX; S4); velocity dispersion (M XII; S3); statistical errors $\sigma(N)$, σ_τ; σ_8	subscript; argument; section
τ	the matrices τ^A_{1644}, τ^A_{1671}; optical depth of reionization, with its prior value τ_0	tag and index; the optical depth occurs in the cosmological sections only
ϕ, φ	expectation value ϕ of the Vorton-Scalar; chiral parts $\phi^{L,R}_{1671}$; shape $\varphi(k)$ of the gap function; phase φ of the order parameter; variable of the zero-dimensional test model	tag and superscript; argument; section
Φ	Φ_{VS}; two-particle content $\Phi_X(\vec{p})$, $\Phi_a(\vec{p})$; the generic field Φ of the general method (M VII A; S4 Sec. 1)	suffix VS; subscript
χ	chirality matrix; the statistic χ^2	exponent 2
ω	ω_{1644}, ω_{1671}; the densities ω_b, ω_c (M XI; S3); a cutoff frequency ω_c in the textbook formula for the gap (M VII; S1 Sec. 5.2)	tag; subscript and section

Seven letters were changed in the documents while this provenance was written, because two meanings occurred close to each other: in the manuscripts the sum $\sum_a |\Delta^a|^2$ is written $|\Delta|^2$ (the supplements call it S, which the manuscripts need for the propagator $\mathcal{S}$); the scattering length is written a_{sc}; the expansion coefficients of the effective potential are written A_2, A_4 (the supplement S4 calls them α, β, the letters of the Hamiltonian matrices); the gradient coefficient is written K_{GL}; the index of the four space axes in the evolution formula of 4+4 is j, not a; the matrix of the mock data in the likelihood is written $\widehat{C}_\ell$, not D_ℓ; and, in the manuscripts and in S1, the generator of an

internal symmetry is written g, not a.

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5. The original files, one by one

For each file: what it is, which conventions it uses, which of its symbols collide with symbols of other files, and what the project does. The provenances P2 (papers) and P3 (notebooks) state how each file was used. Every statement about a notebook in this section and the next was read off the notebook itself; for the symbols, the defining statements were collected by the program of Appendix A.

5.1 The notebooks

Id	Notebook	Symbols defined	of these, also defined in another notebook
N01	4+4.nb	144	142
N02	7+1.nb	1429	86
N03	Bessel product.nb	6	0
N04	Dirac_Einstein-Lovelock_4+4_Nash.nb	80	79
N05	Dirac_Nash_Identity.nb	82	41
N06	Einstein-Lovelock-4+4-Nash.nb	69	69
N07	Einstein-Lovelock-7+1-Nash.nb	68	68
N08	Einstein-Lovelock-Eternal.nb	67	59
N09	Generalized _Kronecker_Delta_4+4.nb	68	55
N10	Pair_Creation_of_Universes_ WaveFunctionOfUniverse-4+4-Einstein-Lovelock-Nash.nb	368	278
N11	WaveFunctionUniverse-4+4-Einstein-Lovelock-Nash.nb	193	174
N12	WaveFunctionUniverse-4+4-Einstein-Lovelock.nb	247	200
N13	det-g.nb	6	5
N14	field_equations.nb	60	60

The numbers in the table come from the scan described in Appendix A.

N01, 4+4.nb (title: “eight, real-valued, 16 by 16, Dirac Matrices”). The source of the 4+4 matrices. *Conventions:* a text at its head gives the coordinates $\{x0; x1, x2, x3; x4; x5, x6, x7\}$ as hidden space, 3-space, time and “superluminal deflating time”; index $A = 0, \ldots, 7$; metric η4488 $= \mathrm{diag}(1, 1, 1, 1, -1, -1, -1, -1)$, space-like $= +1$. With the package Notation the notebook turns typeset expressions into names: τ with an overbar, T16^A, F_A^a. *Symbols taken over:* τ[A] and τ with an overbar $\to \tau^A_{1644}$, $\bar{\tau}^A_{1644}$; σ = σ88 $\to \sigma_{88}$; T16^A[A] $\to \gamma^A_{1644}$; T16^A[8] $\to \omega_{1644}$; P_L, P_R $\to P^{L,R}_{1644}$; σ16 $\to C^{-}_{1644}$; η4488 $\to \eta_{1644}$; s4by4[h], t4by4[h] $\to s_h$, t_h. *Collisions:* γ is a list of sixteen 4×4 matrices (four matrices in N05, a tensor index in N09, the 16×16 Dirac matrices in the project); σ, τ, ζ are matrices here and tensor indices in N09; h is the number 8 that selects a unit spinor (h = hUSE), Ω is the matrix σ.τ[7], J and B are auxiliary. Inside Block[...] the notebook also uses the letters m, ψ, t, b, r, s as temporary variables (m, for example, for a 16×16 array of unknowns); these are local and are not counted as symbols of the notebook. *Remedy:* only the tagged symbols are used; the auxiliary variables are not used at all.

N02, 7+1.nb (title: “7+1: the octonions and the Clifford algebra of 7+1 spacetime”; 27 sections). The source of the 7+1 matrices. The notebook says that it was generated by a script for the author’s 7+1 project and evaluated in Mathematica; it reproduces the objects of N01 under plain names and

builds the 7+1 objects from them. *Conventions:* the same eight coordinate names as N01, but with $x0$ as the time (its Sec. 1.1 says so in a table); two sign conventions side by side (its Sec. 1.2), `η71` $= \mathrm{diag}(-1, 1, \dots, 1)$, which is the convention of N01, and `η17` $= -$`η71`. *Symbols taken over:* `Γ17up[A]` $\to \gamma_{1671}^{\mathrm{swap}_{04}(A)}$; `Γ17[8]`, `ω17` $\to \omega_{1671}$; `χ17` $\to \chi_{1671}$; `PL17`, `PR17` $\to P_{1671}^{L,R}$; `τ0[A]` $\to \tau_{1671}^{\mathrm{swap}_{04}(A)}$; `βD`, `αD[i]` $\to \beta_{1671}$, α_{1671}^{j}; `η17` $\to \eta_{1671}$ after the exchange of the index values 0 and 4; `S44` $\to C_{1644}^{+}$; `dirac16complex` $\to \psi_{1671}$ (dirac1671); `EL`, `ER` $\to E_{1671}^{L,R}$. *Collisions inside the file:* `Γ17` (real, its Sec. 4.2) against `Γ71` (complex, Sec. 4.3) and `Γ71R32` (real 32×32); `η17` against `η71`; `χ17` against `χ71`; `PL17`, `PR17` against `PL71`, `PR71` and `PL44`, `PR44`; `τ` (4+4) against `τ0` (octonionic). Several letters change their meaning from section to section: `H` is the Hamiltonian of the field (Sec. 7.7), a flavour Hamiltonian (Sec. 19) and a constant declared for the tensor package (Sec. 12.2); `M` is an intertwining matrix (Sec. 4.4), the 8×8 matrix $m + p_i$ `τ0[i]` (Sec. 7.4) and a Kohn–Sham mass (Sec. 9.3); `S` is a unitary matrix (Sec. 4.3), a scalar potential (Sec. 8.4) and the action (Secs. 12, 17); `κ` is a chiral density (Sec. 9.1) and the constant $8\pi G_8$ (Sec. 11); `h` is the Hamiltonian matrix (Sec. 7.3) and an induced metric (Secs. 20.3, 25.6); `m` is the mass of the field and, for the tensor package, an abstract index. *Collisions with other files:* `x0` is the time here and a hidden space coordinate in N01 and N04; the dictionary of its Sec. 1.3 renames objects of N01 (Sec. 6.2 below); `sAB` is a sign here and a generator in N10 to N12. *Remedy:* the project uses the real set only, under the tag 1671, with the time at index 4; the relabelling swap_{04} is stated in "dirac matrices", Secs. 0.1 and 2.1, and verified by the checks `71-N1` to `71-N3`; the registry marks the complex set with the word NOT.

N03, `Bessel product.nb` (title: "Bessel Product"). Identities for products of Bessel functions of arguments proportional to $e^{\pm t/6}$, and coefficients of an expansion in such products. It defines six symbols (`cApprox`, `capprox`, `rc`, `rC`, `eqΦ0007`, `sν`), none of which occurs in another notebook or in the project. Not used.

N04, `Dirac_Einstein-Lovelock_4+4_Nash.nb` (title: "Scratch work of Author"). Gravity on the 4+4 spacetime with the tensor package xAct. *Conventions:* `X = {x0[], …, x7[]}`; its text states that $x0$ is a hidden space coordinate (a small circle), $x1, x2, x3$ the usual space coordinates, $x4$ the time and $x5, x6, x7$ "superluminal deflating time" coordinates. *Symbols taken over:* the list `X` and the meaning of the coordinates $\to X = \{x0, \dots, x7\}$ of the project. *Collisions:* `g44` is the metric tensor of a curved 4+4 spacetime (the first version of the project's programs used this name for the flat Dirac matrices: error E7); `H` is a fundamental inverse length of the field equations and `Λ` their cosmological term (in the project `h` is a Hamiltonian matrix, H_0 the Hubble constant and Λ a Lorentz matrix or a cutoff); `β1`, `β2`, `β3` are functions in the metric (in the project β is a matrix); the letters `a`, `b`, `c`, `d`, `f`, `i`, `j`, `k`, `l`, `m`, `n`, `p`, `q`, `r`, `s`, `y` are abstract tensor indices. The name `T16^A` occurs in a text cell only. *Remedy:* the flat metric of the project is η_{1644}; `g44`, `H`, `Λ`, `β1` to `β3` are not used.

N05, `Dirac_Nash_Identity.nb`. Identities of real 4×4 matrices (the subject of the paper D06). *Conventions:* `X = {x1, …, x6}`, six coordinates. *Collisions:* `X` (six coordinates against eight); `γ` (a list of four 4×4 matrices); `σ` is the 4×4 matrix `γ64` here, not the 8×8 matrix of N01 (which N05 defines under the name `σ88` only); `η44` is the 4×4 metric $\mathrm{diag}(1, 1, 1, -1)$; `S44αβ` denotes commutators of the four matrices `γ` (nothing to do with `S44` of N02). `η4488`, `σ88`, `s4by4`, `t4by4` are defined by the same statements as in N01. *Remedy:* nothing is taken from this notebook.

N06, `Einstein-Lovelock-4+4-Nash.nb`; N07, `Einstein-Lovelock-7+1-Nash.nb`; N14, `field_equations.nb`. The Einstein–Lovelock field equations on curved 4+4 and 7+1 spacetimes with xAct. *Conventions:* the list `X` of N04, with the same explanation of the coordinates. *Collisions:* `g44` (N06, N14) and `g71` (N07) are curved metrics. In N06 and N07 a chapter title and a text cell that is not evaluated still show the model metric in an older convention, with a coordinate `x8` in the place of `x0` (the convention of N08); the evaluated cells use `x0[]`. `H`, `Λ`, `β1` to `β3` and the index letters are as in N04. *Remedy:* nothing is taken from these notebooks; the second stage of the author's programme, to which they belong, is not implemented in this project.

N08, `Einstein-Lovelock-Eternal.nb`. A solution of the same field equations. *Conventions:* a chart with the coordinates `x1[]` to `x8[]` and the index values 1 to 8; the metric is called `g`; the coordinate `x8` occupies the place of the hidden coordinate, and `β1`, `β2` are functions of it. The cosmological term enters the vacuum equations with the sign opposite to that of N04, N06, N07 and N14. *Collision:* the coordinate names are shifted with respect to N04 (`x8` here corresponds to `x0` there; `x1` is the

first coordinate here and the second there). *Remedy:* not used; the project's coordinates are those of N04.

N09, `Generalized _Kronecker_Delta_4+4.nb`. The generalized Kronecker delta and the Lovelock tensors with xAct. *Conventions:* chart $x0, \dots, x7$; metric `g` of signature 4+4. *Collisions:* besides the Latin index letters of N04, the Greek letters `α`, `β`, `γ`, `δ`, `ε`, `ζ`, `η`, `κ`, `λ`, `μ`, `ν`, `ξ`, `ρ`, `σ`, `τ`, `χ` are declared as abstract tensor indices; in N01 four of them (`γ`, `ζ`, `σ`, `τ`) are matrices. `g` is a metric here and in N08. *Remedy:* not used.

N10, N11, N12 (the three notebooks on the wave function of the universe; their file names are in the table above; title of each: "Scratch work of Author"). Work on a wave function of the universe on the 4+4 spacetime; each repeats the construction of N01 with the same statements, and N10 also contains the material of N05. *Conventions:* `X = {x0, …, x7}` as plain symbols, with the meaning of N04. *Collisions:* `Ψ16` is a list of sixteen functions of `x0` and `x4`, called the wave function of the universe; `ω` is the array of the coefficients of a spin connection (in the project ω_{1644} is a product of Dirac matrices); `S` is a transformation of spinors, `Γ` the Christoffel symbols; `SAB` and `sAB[A][B]` are the commutators $\frac{1}{4}[$`T16^A[A]`, `T16^A[B]`$]$ (in N02 `sAB` is a sign); `T16^α` are Dirac matrices with a spacetime index; in the text of N10, $\pm M$ are masses of universes and `H` is "some function of M". *Remedy:* not used; the matrices are taken from N01. The commutators `SAB` are the same matrices as Σ^{AB}_{1644} of the project, which defines them independently.

N13, `det-g.nb`. A short calculation of a determinant for a diagonal metric built from `β1`, `β2`, `β3`. Not used.

5.2 The papers

Id	File	Paper
D01	`6dim dirac.pdf`	E. A. Lord, "The Dirac spinor in six dimensions", Proc. Camb. Phil. Soc. 64, 765 (1968)
D02	`A Remarkable Representation.pdf`	P. A. M. Dirac, "A remarkable representation of the 3 + 2 de Sitter group", J. Math. Phys. 4, 901 (1963)
D03	`A new spin-half.pdf`	P. L. Nash, "A new spin-1/2 wave equation", J. Math. Phys. 26, 1363 (1985)
D04	`Bryant-octon-17-2-185-txt.pdf`	R. L. Bryant, "Submanifolds and special structures on the octonians", J. Differential Geom. 17, 185 (1982)
D05	`Calibrated geometries.pdf`	R. Harvey and H. B. Lawson, Jr., "Calibrated geometries", Acta Math. 148, 47 (1982)
D06	`Identities satisfied.pdf`	P. L. Nash, "Identities satisfied by the generators of the Dirac algebra", J. Math. Phys. 25, 204 (1984)
D07	`On the structure of the split octonion algebra.pdf`	P. L. Nash, "On the structure of the split octonion algebra", Nuovo Cimento B 105, 31 (1990)
D08	`Second gravity.pdf`	P. L. Nash, "Second gravity", J. Math. Phys. 51, 042501 (2010)
D09	`Vector cross products.pdf`	R. B. Brown and A. Gray, "Vector cross products", Comment. Math. Helv. 42, 222 (1967)

D07 and D08 define real 8×8 matrices τ^A, $\bar\tau^A$ (there called reduced Brauer–Weyl generators) of the kind from which the 4+4 Dirac matrices of N01 are built, and D08 defines the 16×16 matrices t^A with the blocks $(0, \bar\tau^A; \tau^A, 0)$. *Conventions that differ from the notebooks:* the index runs over $A = 1, \dots, 8$, with $\tau^8 = \bar\tau^8 = 1$; in N01 and in the project it runs over $A = 0, \dots, 7$, with $\tau^0 = \bar\tau^0 = 1$. The metric is called G_{AB}, with the diagonal $(1, 1, 1, -1, -1, -1, -1, 1)$ for $A = 1, \dots, 8$. A tilde denotes the transpose (in the project a superscript T does, and $\tilde\psi$ is an adjoint). D07 adds that, relative to an earlier paper of the same author, "tau bar" and "tau" are exchanged and the sign of τ^7 is changed; and it uses the letter H for a product of the seven matrices with $A = 1, \dots, 7$, which it requires to be $+1$ (in the scanned copy the formula is not legible enough to say which of the factors carry a bar). *Remedy:* the project takes its matrices from N01, not from the papers, and uses τ^A_{1644},

$\bar{\tau}^A_{1644}$, γ^A_{1644}, η_{1644} with $A = 0, \dots, 7$; a formula of D07 or D08 with the index value 8 corresponds to the index value 0 of the project. For the matrices of the project, $\tau^0_{1644} = 1$ and $\tau^1_{1644} \cdots \tau^7_{1644} = 1$ (check `44-B1`). They were not compared entry by entry with the matrices printed in the papers.

D06 treats fifteen real 4×4 matrices γ^{AB}, labelled by a pair of indices with the values 1 to 6 and a metric of signature 3+3, and calls the six antisymmetric ones s^h (self-dual) and t^h (anti-self-dual), $h = 1, 2, 3$. The notebooks N01 and N05 follow it: their `s4by4[h]`, `t4by4[h]` carry these letters, and the list `γ` of N01 consists of sixteen matrices of this kind (the six, their nine products `s4by4[J].t4by4[K]` and the unit matrix). *Remedy:* the project keeps the letters s_h, t_h for the 4×4 blocks and never writes γ without two things: an upper index with the values 0 to 7, and the tag of the spacetime where the matrix is defined.

D01 (Lord) gives the general procedure by which D07 and D08 build spinor representations from reduced generators; it works with the Dirac matrices of four dimensions and with spinors of five and six dimensions. **D03** derives a wave equation for a massive particle of spin $\frac{1}{2}$ from a Casimir operator, and **D02** constructs a representation of the 3+2 de Sitter group in terms of two oscillators. In these three papers γ, σ, ψ and the indices have four-, five- or six-dimensional meanings. *Remedy:* none of their symbols is used.

D04, D05 and D09 (Bryant; Harvey and Lawson; Brown and Gray) are the mathematical literature on the octonions, on which the 7+1 construction rests. The project quotes them for the existence and the properties of the octonion algebra and takes no formula in their notation; its octonionic matrices are τ^A_{1671}, defined in its own documents and verified by machine (checks `71-B1`, `71-D6`). The copies of these papers are scans, and the text extracted from them does not render their Greek letters reliably; their notation was therefore not tabulated. One collision is visible even so: D09 writes X for a vector cross product, the letter that the notebooks use for the list of coordinates.

5.3 The project NCSS

The task specification asks that the author's project NCSS ("Neutrino Condensate Singlet Superfluid", github.com/once-ere/NCSS, folder `Revision_1`) be read for context. Three things were taken from it: the name Vorton-Scalar and its meaning; two data files with spectra of the standard cosmological model, which serve as the reference in the validation of the Boltzmann solver (`data/reference/`, with checksums); and the scripts that build the documents and the manuscripts, which were adapted (`build_docs.sh`, `build_manuscripts.py`, `tex2md.py`; each says so in its header). NCSS has a symbol table of its own (its Supplement S4). The symbols that matter here:

NCSS	Meaning in NCSS	In this project
χ_V, also χ	the Vorton-Scalar: a *real* scalar field	Φ_{VS}, a *complex* scalar field; χ is the chirality matrix
m_χ	mass of the Vorton-Scalar	m_s, mass of the radial mode; the phase mode ϑ_{VS} is massless
g, g_{ij}; $\alpha = g^2/4\pi$	coupling to the neutrinos (a matrix in the space of the three species)	$g_{VS} = m/v_{VS}$; $\alpha_{VS} = g^2_{VS}/4\pi$
D, D_c	$D = \alpha m_\nu / 2m_\chi$; two neutrinos bind for $D > D_c = 0.83990$	$D_{VS} = \alpha_{VS} m/m_s$, **twice** the combination of NCSS; two fermions bind for $D_{VS} > D_c = 1.67981$
C	$g^2 m_\nu^2/m_\chi^2$, the "nugget parameter"; also the charge-conjugation matrix of four dimensions	$C = g^2_{VS} m^2/m_s^2$, the same combination for the fields of this project; the matrices are $C^{\mp}_{1644}$, $C^{\pm}_{1671}$
ϕ	$\langle \chi_V \rangle$	ϕ with $\langle \Phi_{VS} \rangle = \phi/\sqrt{2}$
Δ_{ij}	gap matrix in the space of the three neutrino species	Δ^a, $a \in \{0, 5, 6, 7\}$, a vector along the four extra axes
ν_i	the neutrino fields	ψ_{1644}, ψ_{1671}; ν is the number 8 of internal states
S	a stability parameter of neutrino matter	Sec. 4
λ	$(D/\pi x_F)\ln(1 + 4x_F^2)$, with $x_F = k_F/m_\chi$	$\lambda = N(0)\, W(k_F, k_F)$; x is m^*/m
metric	four dimensions, signature $(+,-,-,-)$	eight dimensions: η_{1644} (space-like $= +1$), η_{1671} (time-like $= +1$)

Two warnings follow from the table. A value of D must not be carried from one project to the other without the factor 2. And the thresholds of the coupling C quoted in the two projects are not comparable: in NCSS the scalar is real with a mass term and each species has two internal states, here the scalar is complex with a symmetry-breaking potential and the field has eight internal states (Supplement S4, Sec. 5). *Remedy:* every quantity of the Vorton-Scalar of this project carries the suffix VS in the registry, so that it cannot be mistaken for the quantity of NCSS with the same letter.

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6. Collisions between the original files, symbol by symbol

6.1 The same symbol for different objects

The table lists the symbols of the original files that (i) have different meanings in different files and (ii) concern an object that the project uses. The complete machine-made list of the symbols that several notebooks define is in Appendix A.

Symbol	Meaning, by file	In the project
`X`, `x0` … `x7`	N01, N04, N06, N07, N10–N12: coordinates $x0, \dots, x7$, with the statement that $x0$ is a hidden space coordinate and $x4$ the time. N09, N14: the same chart without explanation. N02: the same names, $x0$ time. N05: `X` $= \{x1, \dots, x6\}$. N08: coordinates $x1, \dots, x8$, $x8$ hidden	$X = \{x0, \dots, x7\}$ with the meaning of N01 and N04, never renamed; N02 by swap_{04}
`x8`	N08: the hidden coordinate. N06, N07: in a chapter title and a text cell, where the evaluated cells have `x0`	not used
`g44`, `g71`, `g`	N04, N06, N14: `g44`, curved metric of signature 4+4. N07 and N02 (Sec. 12.2): `g71`, curved metric of signature 7+1. N08, N09: `g`, curved metric	the flat metrics are η_{1644}, η_{1671}; `g44`, `g71` never denote project objects
`η17`, `η71`	N02: two metrics of the 7+1 signature, `η17` $= -$`η71`	η_{1671} is `η17` with the index values 0 and 4 exchanged
`η44`	N05, N10: the 4×4 metric $\text{diag}(1, 1, 1, -1)$ (in N05 through a symbol `g_αβ`)	not used; the error E7 concerned names of this kind
`γ`	N01, N10–N12: sixteen 4×4 matrices. N05: four 4×4 matrices. N09: tensor index. D06: γ^{AB}, fifteen 4×4 matrices	γ^A_{1644}, γ^A_{1671}: real 16×16 Dirac matrices
`Γ17`, `Γ71`, `Γ71R32`; `Γ`	N02: real 16×16, complex 16×16 and real 32×32 generators of 7+1 Clifford algebras. N10–N12: `Γ`, Christoffel symbols	γ^A_{1671} from `Γ17up`; the other two sets are not used
`τ`, `ζ`, `τ0`	N01, N02, N10–N12: `τ[A]` (and `ζ[A]`), $A = 0, \dots, 7$, the 8×8 blocks of 4+4. N02: `τ0[A]`, the octonionic matrices. N09: `τ`, `ζ` tensor indices. D07, D08: τ^A, $A = 1, \dots, 8$	τ^A_{1644} and τ^A_{1671}; the plain letter τ is the optical depth
`σ`, `σ8`	N01, N02, N10–N12: `σ`, the 8×8 matrix $(0, 1_4; 1_4, 0)$. N05: `σ` $=$ `σ8` $=$ `γ64`, a 4×4 matrix. N09: tensor index. D08: σ lowers spinor indices. N02 (Secs. 7.11, 9.1): σ^A, $\bar{\sigma}^A$, a family of 8×8 matrices	σ_{88}; other uses of σ in Sec. 4
`S44`, `S44αβ`	N02: `S44`, the product of the four time-like 4+4 Dirac matrices. N05, N10: `S44αβ`, commutators of four 4×4 matrices	C^+_{1644}
`sAB`, `SAB`	N10–N12: the commutators $\frac{1}{4}$[`T16^A[A]`,`T16^A[B]`]. N02: `sAB`, a sign	Σ^{AB}_{1644}
`ω`, `ω17`	N10–N12: `ω`, coefficients of a spin connection. N02: `ω17` $=$ `Γ17[8]`; in its text ω is also a frequency	ω_{1644}, ω_{1671}: products of the eight Dirac matrices
`χ17`, `χ71`; `χ`	N02: chirality matrices of the real and of the complex set; in the text of its Sec. 13.2, χ is a rescaled field. N09: `χ` tensor index. NCSS: χ is the Vorton-Scalar	χ_{1671} from `χ17`
`P_L`, `P_R`, `PL…`, `PR…`	N01, N10–N12: `P_L`, `P_R` (4+4). N02: `PL44`, `PR44` (4+4); `PL17`, `PR17` (real 7+1 set); `PL71`, `PR71` (complex set)	$P^{L,R}_{1644}$, $P^{L,R}_{1671}$
`Ψ16`, `ψ`, `Ψ`	N10–N12: `Ψ16`, sixteen functions of `x0`, `x4` (wave function of the universe). N02: ψ, the field `dirac16complex`; `Ψ`, a pair of octonions	ψ_{1644}, ψ_{1671} (dirac1644, dirac1671)

Symbol	Meaning, by file	In the project
H	N04, N06, N07, N08, N14 (and N02, Sec. 12.2): a fundamental inverse length, declared as a constant. N02: Hamiltonian (Sec. 7.7), flavour Hamiltonian (Sec. 19). N10: a function of a mass M. D07: the product $\tau^1 \cdots \tau^7$	h_{1644}, h_{1671} for the Hamiltonian matrices; H_0 for the Hubble constant
Λ	N04, N06, N07, N08, N14: cosmological term (with opposite signs in N08 and in the others). N02: Lorentz matrix (Sec. 5.1), cutoff (Sec. 9.6)	Lorentz matrix or cutoff (Sec. 4)
β1, β2, β3	N04, N06, N07, N13, N14: functions of x0[] and x4[] in a metric. N08: functions of x8[] and x4[]. N10–N12: the same with plain x0, x4	not used; β_{1644}, β_{1671} are matrices
index letters	N02, N04, N06–N09, N14: a, b, c, d, f, i, j, k, l, m, n, p, q, r, s, y are abstract tensor indices; N09 adds sixteen Greek letters. D07, D08: $A = 1, \dots, 8$. D06: $A, B = 1, \dots, 6$; $h = 1, 2, 3$	$A, B = 0, \dots, 7$; $a, b \in \{0, 5, 6, 7\}$; i, j space-like; $t = 5, 6, 7$; $h = 1, 2, 3$ (Sec. 3.1)

6.2 Different symbols for the same object

The converse also occurs: the same object carries different names in different files. The dictionary of 7+1.nb (its Sec. 1.3) lists the renamings that N02 made; the table adds the papers and the names of the project.

Object	N01 (and N10–N12)	N02	Papers	Project
8×8 blocks	τ[A], τ with an overbar	τ[A], τbar[A]	D07, D08: τ^A, $\bar{\tau}^A$ (A shifted)	τ^A_{1644}, $\bar{\tau}^A_{1644}$
4+4 Dirac matrices	T16^A[A]	T16[A]	D08: t^A	γ^A_{1644}
their product	T16^A[8]	T16[8]		ω_{1644}
chiral projectors of 4+4	P_L, P_R	PL44, PR44		$P^{L,R}_{1644}$
metric of 4+4	η4488, η8, g_AB	η4488, gAB44	D08: G_{AB} (A shifted)	η_{1644}
product of the four space-like matrices	σ16	σ16		C^{-}_{1644}
product of the four time-like matrices		S44		C^{+}_{1644}
commutators	SAB, sAB[A][B] (N10–N12)			Σ^{AB}_{1644}
4×4 blocks	s4by4[h], t4by4[h]	s4by4[h], t4by4[h]	D06: s^h, t^h	s_h, t_h
the 8×8 matrix $(0, 1_4; 1_4, 0)$	σ, σ88	σ	D08: σ	σ_{88}

6.3 Plain letters of the project that coincide with a symbol of a notebook

A tagged symbol cannot coincide with a symbol of a notebook (check S6). A plain letter can. The following table lists every plain symbol of the registry of Sec. 3 that is, as a string of letters, also a symbol of a notebook (33 symbols), with both meanings. It is generated by the script that performs the checks, and check S9 fails if a coincidence is missing from it. The section numbers are those of 7+1.nb (N02).

Symbol	In the notebooks	In this project
a	N02, N04, N06–N09, N14: abstract tensor index of the manifold M8 (package xAct)	index of the four extra axes, values 0, 5, 6, 7
B	N01, N10: B = 1, an index value in a check; N02: index with the values 0 to 7	index with the values 0 to 7; bag constant (S4, M IX); Killing form $B(X, Y)$ (DM, M II)
b	as a	as a

Symbol	In the notebooks	In this project
`C`	N02: index with the values 0 to 7	index with the values 0 to 7; the coupling $C = g_{VS}^2 m^2/m_s^2$
`h`	N01, N10: `h = hUSE`, the number 8 that selects a unit spinor; N02: Hamiltonian (Sec. 7.3), flavour Hamiltonian (Sec. 19), induced metric (Secs. 20.3, 25.6)	index of the matrices s_h, t_h; Hubble constant in units of 100 km s^{-1} Mpc^{-1} (the Hamiltonian matrix is `h`)
`i`	N02, N04, N06–N09, N14: abstract tensor index (xAct)	index of space-like coordinates
`j`	N02, N04, N06–N09, N14: abstract tensor index (xAct); text of N10–N12: the number 134 of a basis matrix	index of space-like coordinates
`M`	N02: an intertwining matrix (Sec. 4.4), the 8×8 matrix $m + p_i$ `τ0[i]` (Sec. 7.4), a Kohn–Sham mass (Sec. 9.3)	mass of an exchanged boson (the matrix of N02, Sec. 7.4, is M_{1671}; the gap matrix is `M`)
`m`	N02, N04, N06–N09, N14: abstract tensor index (xAct); in the text of N02 also the mass of the field	mass of the fermion field, m_{1644} or m_{1671} with the tag dropped
`P`	N02: the 32×32 matrix $\sum_A p^A$ `Γ71R32[A]` (Sec. 5.5), a pseudoscalar potential (Sec. 8.4)	generator of $U(1)_E$; pressure; thermal boson loop $P(M)$
`Q`	N02 (Secs. 1.2, 1.4): the quadratic form of a Clifford algebra	thermal part $Q(M)$ of a boson loop; electric charge (M X)
`S`	N02: a unitary matrix (Sec. 4.3), a scalar potential (Sec. 8.4), the action (Secs. 12, 17); N10–N12: the transformation $\exp(\frac{1}{2}\omega$ `SAB`$)$ of spinors	element of the spin group (the meaning of N10–N12); classical action $S[\Phi]$; entropy; in S1 and S4 also $\sum_a \lvert\Delta^a\rvert^2$
`U`	N02 (Sec. 20.2): a stabilizing potential	the matrix $U(\vec{p})$ of positive-energy spinors
`V`	N02 (Secs. 8.4, 9.6): a vector potential of the Kohn–Sham theory	effective potential $V[\Delta]$
`W`	N02: the unitary matrix (`EL`, `ER`) (Sec. 7.11); a function $a^3 b^4 U$ (Sec. 20.2)	kernel $W(k, k')$ of the gap equation; generating functional $W[J, K]$ (the grading is W_{1644})
`x`	N02 (Sec. 1.2): an octonion $x = x0\, e_0 + \cdots + x7\, e_7$	the ratio m^*/m
`β`	N09: abstract tensor index; N02: rapidity of a boost (Sec. 7.9)	the matrix β_{1644} or β_{1671} with the tag dropped
`Γ`	N10–N12: the Christoffel symbols returned by a function `rt`	effective action $\Gamma[\phi, G]$; in M VI and S1 a vertex matrix
`ε`	N09: abstract tensor index; N02: the matrix $(0, 1; -1, 0)$ (Sec. 4), an energy density (Sec. 8.7)	energy density $\varepsilon(k_F, x)$
`η`	N09: abstract tensor index; N02: short for `η17`	eigenvalue $\eta(T)$ of the linearized gap equation (the metrics carry a tag or indices)
`κ`	N09: abstract tensor index; N02: a chiral density (Sec. 9.1), the constant $8\pi G_8$ (Sec. 11)	pair correlation $\kappa(k)$
`Λ`	N04, N06–N08, N14: cosmological term of the field equations (xAct constant); N02: Lorentz matrix (Sec. 5.1), momentum cutoff (Sec. 9.6)	Lorentz matrix $\Lambda^A{}_B$; momentum cutoff
`λ`	N09: abstract tensor index; N02: coordinate index	dimensionless pairing coupling
`μ`	N09: abstract tensor index; N02: coordinate index; chemical potential (Sec. 8.7)	chemical potential
`ν`	N09: abstract tensor index; N02: coordinate index	the number of internal states, $\nu = 8$
`σ`	N01, N02, N10–N12: the 8×8 matrix that is σ_{88} here; N05: the 4×4 matrix `γ64`; N09: abstract tensor index	self-energy $\sigma(p)$ (the matrix is always written σ_{88})
`τ`	N01, N02, N10–N12: the matrices `τ[A]`; N09: abstract tensor index	optical depth of reionization (the matrices are always written τ^A_{1644}, τ^A_{1671})
`φ`	N02 (Sec. 20.3): the canonical radion	expectation value of the Vorton-Scalar
`χ`	N09: abstract tensor index; N02 (Sec. 13.2): a rescaled field	chirality matrix of the spacetime at hand
`ψ`	N02: the field `dirac16complex` in formulas	the fermion field (ψ_{1644} or ψ_{1671} where it is defined)
`E_x`	N02 (Sec. 8.4): exchange energy of its Kohn–Sham theory	exchange (Fock) energy of the two modes of the Vorton-Scalar: the same kind of quantity in another model

Symbol	In the notebooks	In this project
`m_s`	N02 (Sec. 17.4): mass of the strange quark	mass of the radial mode of the Vorton-Scalar (name in files: `msVS`)
`λ_P`	N02 (Sec. 24.3): coupling of a Higgs portal	the part of the pairing coupling that comes from the phase mode

In none of these cases does the project take the object of the notebook under the plain letter: where it takes the object (the matrices `σ`, `τ[A]`, the field), it uses the tagged symbol. Two coincidences deserve attention because the letters carry a subscript and look specific: m_s is the mass of the strange quark in N02 and the mass of the radial mode of the Vorton-Scalar here (name in files: `msVS`); and λ_P is a coupling of a Higgs portal in N02 and a part of the pairing coupling here.

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7. Collisions that the project produced itself, and their removal

The project is not exempt from the problem it is asked to solve. Four collisions or errors of this kind were made and removed; each is in the error log (`provenance/P9_error_log`).

1. **E7.** The first version of the programs called the project's matrices `g44`, `g71`, `eta44`, `eta71`, `gamma44`, `gamma71`, `chi71`. The notebooks use `g44` and `g71` for curved metrics, and `η71`, `Γ71`, `χ71` for the *complex* 7+1 set. All names were replaced by the tagged names of Sec. 3; check `S8` verifies that the old names survive only where a notebook is quoted.
2. **E35.** The output of the program used the letters A and B both for the two factors of the internal symmetry of the zero-mode sector and for the two factors of the rotation group of the four space coordinates of 4+4. These are different groups. The second pair is now called $SU(2)_X$, $SU(2)_Y$.
3. **E41.** The text of check `44-D6` defined `alpha1644^a` as $\gamma^4_{1644}\gamma^a_{1644}$, with the sign opposite to the definition $\alpha^j_{1644} = -\gamma^4_{1644}\gamma^j_{1644}$ of S1, of the manuscripts and of the program itself (`kernel.rs`). The statement that the check verifies is true for either sign; the text was corrected.
4. **E42.** The first draft of this provenance was written partly from working notes, and the first version of the program that scans the notebooks recorded only definitions at the beginning of a line. Checking every statement against the notebooks showed that both were wrong in several places; the scanner was rewritten and the statements corrected before anything was committed.

Section 4 records seven letters that were changed in the documents for the same reason.

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8. The machine checks

Check	Statement	Result
`S1`	every name of the registry belongs to exactly one entry (78 names in 113 entries)	passed
`S2`	no name of the registry is a symbol defined in any of the 14 notebooks (1921 notebook symbols compared; exempt: the 8 coordinate names x0 … x7, taken from N04 with the same meaning)	passed
`S3`	65 of the 78 names of the registry carry the tag 1644 or 1671 or the suffix VS; of the other 13, 8 are the coordinate names (deliberately those of the notebooks, S2) and 5 (F_pseudoscalar, F_ scalar, J5, sigma88, swap04) are symbols of no notebook	passed
`S4`	every name with the tag 1644 or 1671 that occurs in a data file or a document is in the registry (48 different names in 94 files; 4 further tokens are names of columns, bibliography keys or program variables)	passed
`S5`	every name with the suffix VS that occurs in a data file or a document is in the registry (17 different names)	passed

Check	Statement	Result
S6	no symbol of any notebook contains the tag 1644 or 1671 or ends with VS, so a tagged name cannot be mistaken for a symbol of a notebook	passed
S7	the manuscripts contain none of the 34 original names of the objects taken from the notebooks (T16, Γ17, η4488, χ17, τO, g44, ...): they use the symbols of the registry only	passed
S8	the names g44, g71, eta44, eta71, gamma44, gamma71, chi71, which the first version of the programs used for project matrices although the notebooks use them for other objects (error E7), occur in the programs and data files only where a notebook is quoted	passed
S9	33 plain (untagged) symbols of the registry coincide, as strings of letters, with a symbol of a notebook (B, C, E_x, M, P, Q, S, U, V, W, a, b, h, i, j, m, m_s, x, Γ, Λ, β, ε, η, κ, λ, λ_P, μ, ν, σ, τ, φ, χ, ψ); every one of them is listed with both meanings (table of Sec. 6.3 of the MASTER provenance), and no listed coincidence is missing	passed

The complete output is the file `data/symbol_checks.txt`:

```
PROJECT spacetimes: checks of the symbol registry (provenance/symbols.csv) against the notebooks
 ↪  and the documents
9 of 9 checks passed

[PASS] S1  every name of the registry belongs to exactly one entry (78 names in 113 entries)
[PASS] S2  no name of the registry is a symbol defined in any of the 14 notebooks (1921 notebook
 ↪  symbols compared; exempt: the 8 coordinate names x0 ... x7, taken from N04 with the same
 ↪  meaning)
[PASS] S3  65 of the 78 names of the registry carry the tag 1644 or 1671 or the suffix VS; of the
 ↪  other 13, 8 are the coordinate names (deliberately those of the notebooks, S2) and 5
 ↪  (F_pseudoscalar, F_scalar, J5, sigma88, swap04) are symbols of no notebook
[PASS] S4  every name with the tag 1644 or 1671 that occurs in a data file or a document is in the
 ↪  registry (48 different names in 94 files; 4 further tokens are names of columns, bibliography
 ↪  keys or program variables)
[PASS] S5  every name with the suffix VS that occurs in a data file or a document is in the
 ↪  registry (17 different names)
[PASS] S6  no symbol of any notebook contains the tag 1644 or 1671 or ends with VS, so a tagged
 ↪  name cannot be mistaken for a symbol of a notebook
[PASS] S7  the manuscripts contain none of the 34 original names of the objects taken from the
 ↪  notebooks (T16, Γ17, η4488, χ17, τO, g44, ...): they use the symbols of the registry only
[PASS] S8  the names g44, g71, eta44, eta71, gamma44, gamma71, chi71, which the first version of
 ↪  the programs used for project matrices although the notebooks use them for other objects
 ↪  (error E7), occur in the programs and data files only where a notebook is quoted
[PASS] S9  33 plain (untagged) symbols of the registry coincide, as strings of letters, with a
 ↪  symbol of a notebook (B, C, E_x, M, P, Q, S, U, V, W, a, b, h, i, j, m, m_s, x, Γ, Λ, β, ε, η,
 ↪  κ, λ, λ_P, μ, ν, σ, τ, φ, χ, ψ); every one of them is listed with both meanings (table of Sec.
 ↪  6.3 of the MASTER provenance), and no listed coincidence is missing
```

What these checks do not prove: that every *sentence* of the documents uses the symbols correctly, and that the hand-written statements of Secs. 4 to 6 are complete. The first is a matter of reading; for the second, the two machine-made lists (Appendix A for the notebooks, `data/symbol_concordance.csv` for the documents of the project) let a reader test any statement.

Appendix A. The symbols that the notebooks define

Method. Each notebook is converted to text, cell by cell, by `tools/nb2txt.py`; typeset input appears in linear form, a subscript as `P_L`, a superscript as `T16^A`, an overbar as `Over[τ,_]`. The script `tools/scan_notebook_symbols.py` then records the symbols that a notebook defines:

- in **input cells**, after comments and strings have been removed, the head of the left-hand side of *every* assignment (`=`, `:=`, `^=`, `^:=`), wherever it stands: at the start of a line, inside parentheses, inside `Do[...]` or `Table[...]`, or in a chain `a = b = c`. Local variables of `Module`, `With`, `Block`

are not counted, nor are the limits of typeset sums, properties of built-in functions and settings whose names begin with `$`;
- the **declarations of xAct**: the manifold and its abstract indices (`DefManifold`), the chart and its coordinates (`DefChart`), the metric and its covariant derivative (`DefMetric`), constants and scalar functions;
- in **text cells and headings**, the symbol that stands immediately to the left of a single "=" in a formula of the prose, because this is notation that a reader of the notebook meets. A symbol that a notebook introduces *only* in this way is marked with an asterisk. This rule has limits, which the header of the script states: a product written without a blank is read as one name, and a bare word of four or more lower-case letters is taken as prose.

A notebook can define a symbol several times with different meanings; the scan keeps the first definition (a definition in code before one in prose), and Sec. 5 treats such cases by hand. Output and message cells are ignored. The defining statements themselves are not written into the repository, because the notebooks are unpublished; the option `--dump` of the script writes them to a file for checking.

Result. 1921 symbols; 350 are defined in two or more notebooks. The table groups these by the set of notebooks that define them (54 sets). "Same defining text" means that the first defining statement, with blanks removed, is identical in all notebooks of the set; different text does not prove that the objects differ, and identical text does not prove that they agree (the right-hand sides may contain symbols that differ). The column is a guide: 274 symbols have the same text, 76 have not, and Secs. 5 and 6 treat every symbol that matters for the project individually. The complete list, with the symbols that only one notebook defines, is the file `provenance/generated/notebook_symbols.csv`.

Notebooks	Symbols with the same defining text	Symbols with different defining text
N02 N04 N06 N07 N08 N09 N10 N11 N12 N14		`j*`, `x4*`, `x5*`, `x6*`, `x7*`
N02 N04 N05 N06 N07 N10 N11 N12 N14		`X`
N02 N04 N06 N07 N09 N10 N11 N12 N14		`x0*`
N04 N06 N07 N08 N09 N10 N11 N12 N14	`dir`, `header`, `name`, `names`	
N04 N06 N07 N08 N10 N11 N12 N13 N14		`β1`, `β2`, `β3`
N02 N04 N06 N07 N08 N09 N13 N14	`kδ`	
N02 N04 N06 N07 N10 N11 N12 N14		`MyArrayComponents*`
N04 N06 N07 N08 N09 N10 N11 N12	`time1`	
N04 N06 N07 N08 N10 N11 N12 N14	`A_a^b*`, `DIM8`	
N01 N02 N05 N09 N10 N11 N12		`γ*`, `σ`

Notebooks	Symbols with the same defining text	Symbols with different defining text
N02 N04 N06 N07 N08 N09 N14	CD, M8, a, b, c, chartM8, d, f, i, k, l, m, n, p, q, r, s, y	x1, x2, x3
N02 N04 N06 N07 N08 N10 N14		H*
N04 N06 N07 N10 N11 N12 N14	sX0, ssX	
N01 N02 N05 N10 N11 N12	ID4, ID8, Qa, Qb, SelfDualAntiSymmetric, Zero4, Zero8, evalues, evecs, s4by4, t4by4	AntiSelfDualAntiSymmetric, η4488
N01 N02 N09 N10 N11 N12		τ
N02 N04 N06 N07 N08 N14		Λ*
N04 N06 N07 N08 N09 N14	dimM8	indices
N01 N02 N09 N11 N12		ζ
N01 N02 N10 N11 N12	Zero16	ID16, Ω*, σ16
N01 N04 N10 N11 N12		T16^A*
N01 N05 N10 N11 N12	ε3, ε4	
N02 N04 N06 N07 N08	EinsteinCD, RicciCD, RicciScalarCD, RiemannCD	
N02 N05 N10 N11 N12		u*
N02 N06 N07 N08 N09		Lovelock1*, Lovelock2*, Lovelock3*
N04 N06 N07 N08 N14	org	EinsteinLovelockVacuumFieldEquations, allLovelock1, allLovelock2, allLovelock3
N04 N06 N10 N11 N12		MatrixMetric44
N01 N10 N11 N12	BASE16, G16, Id16, MId, MId16, Over[τ,_], P_L, P_R, Simp, Simp16, ZERO16, anti, bas64, base16, determineIF8AntiSymmetric, dups, dupssymm16, find8AntiSymmetric, mid, simp, st, symm, symm16, t160, t16A, t16AB, t16ABC, t16ABCD, t16ABCDE, t16ABCDEF, t16ABCDEFG, t16ABCDEFGH, tA, tAB, tABC, use16, η64, σΩ	
N02 N10 N11 N12		S*, sAB, v*, ω*
N04 N06 N07 N08	eqsELei00	almightyWs
N04 N06 N07 N14	My4ArrayComponents, TableDelta	TableMetric
N04 N10 N11 N12	eextract, makeAntiSym, makeSym, skelx	
N06 N07 N11 N12	tau16*	
N01 N02 N10	EA, hUSE	B*, J*, g_AB*, h*, unit
N01 N05 N10	Over[σ22,_], anti4by4, symm4by4, txA, txA0, txAB, txABC, η16, η2244, σ22	txABCD
N01 N10 N12	coupled, coupled1, coupled2, linTrans, linTransAntiSymmetric, linTransSymmetric, negativeTrMM, positiveTrMM	

Notebooks	Symbols with the same defining text	Symbols with different defining text
N02 N08 N09		g*
N04 N05 N10	MergeSetsStep, dualRawSets, showCoupledEquations	rawSets
N04 N06 N07	ein	
N04 N06 N14	g44	
N04 N08 N14		Lovelock1a, Lovelock2a
N04 N10 N12		blocks
N04 N13 N14	test0	
N10 N11 N12	Au*, Ax*, E_((A))^α*, EinsteinG, LagrangianΨ16*, Lj, Lsquared*, RS, RicciΓ, RieΓ, SAB, SAB1, SAB2, T16^α, T^α*, Universe*, ass, con, constraintTrig, constraintVars, constraintX, covariantDiffMatrix, detgg, eAa, eAa04, eL, eLa, e_((A))^α, e_α^((A)), eηe, g4488, g_(μν|α)*, ginv, gtry, gtrye_((A))^α, gtrye_α^((A)), iid, makeSubstitution, qid, result, rt, sfψ16Aa, sg, sge_((A))^α, sge_α^((A)), sgtrye_((A))^α, sgtrye_α^((A)), spinCoefficients, spinCoeffs, sreplaceZ, ssgm4488, subsDefects, sx0x4, szt, sω, sωf, sωfgtry, sωgtry, useDSQRT, useT16, usegT16, usingLagrangianF16massive*, Γ, Γ_j*, Ψ16, Ψ16lower, Ψ16upper, Ω16, η99, ωmat, ωs, ωμIJ, ωω	La, Lg, TU^μν*, eLazt, imes
N01 N05	σ88	
N01 N10	B1*, E^A_a, E^a_A, F_A^a, F_a^A, JT4, Jcomplement, JcomplementProjection8, Jcomplementcc, MS1S2, Myid, T4cc, T7cc, \[DoubleStruckU], antisymm16, base4by4, basisProjections16, basisProjections8, caΨccΨ, caΨccΨ2, eqsRQQ7Q8, evaluesAB, evecsAB, findComplexStucture*, gridx, jjcc, myid, parallel04, possibleComplexStructureIndex, possibleComplexStructureIndex2, possibleComplexStructureMatrix, possibleComplexT4StructureIndex, projection04, realProjection16, realProjection8, sJu, sQ7, sQ7Q8, sQ8, sRQQ7Q8, seqsRQQ7Q8, sixers, sixersIndices, sixersJ, type1SpinorBasisMatrix, type2SpinorBasisMatrix, unit3, xxx, yyy, Ψ16cc, Ψ16ccA, Ψ16cccc, τABC	
N02 N07	g71	
N02 N09		α*, β*, δ*, ε*, η*, κ*, λ*, μ*, ν*, ξ*, ρ*, χ*
N04 N10	blockPartition	
N04 N14	gg44	makeSym00
N05 N10	S44αβ, g3, g_αβ, sε, γABs	η44
N06 N07	MatrixMetric71, a4	
N08 N14		Lovelock3a
N10 N12	ConvertMapleToMathematica, GetTokens, IsAlpha, IsDigit, IsSpace, MapleToMathematicaFunction, ParseTokens, ToMathematicaString, helpf, mapleztStringToMathematica, snewfψ16Aa, sreplacenZ, sta, sti, times, ω_((B)(A))*	ProcessASTNode
N11 N12	LagrangianF16massive*	matTest, resolveWithRemainder